Exam Date & Time: 22-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

COMPOSITE STRUCTURES

Composite Structures [AAE 4050]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions. Section Duration: 180 mins

Answer all questions.
Assume any missing data with proper reasoning.

1A) Describe three major advantages of using composite structures in engineering applications and provide brief examples to illustrate each advantage. (3)

1B) Discuss the various types of matrices commonly employed in composite materials, providing a brief description of each type along with a specific application in the field of composites. (3)

1C) Assuming that the fibres in a unidirectional continuous fibre composite are arranged in a hexagonal packing as shown in figure 1, show that the fibre volume fraction is given by $3.626\ (r_f/a)^2$. Determine the maximum volume fraction of fibres that can be packed in this fashion. (4)

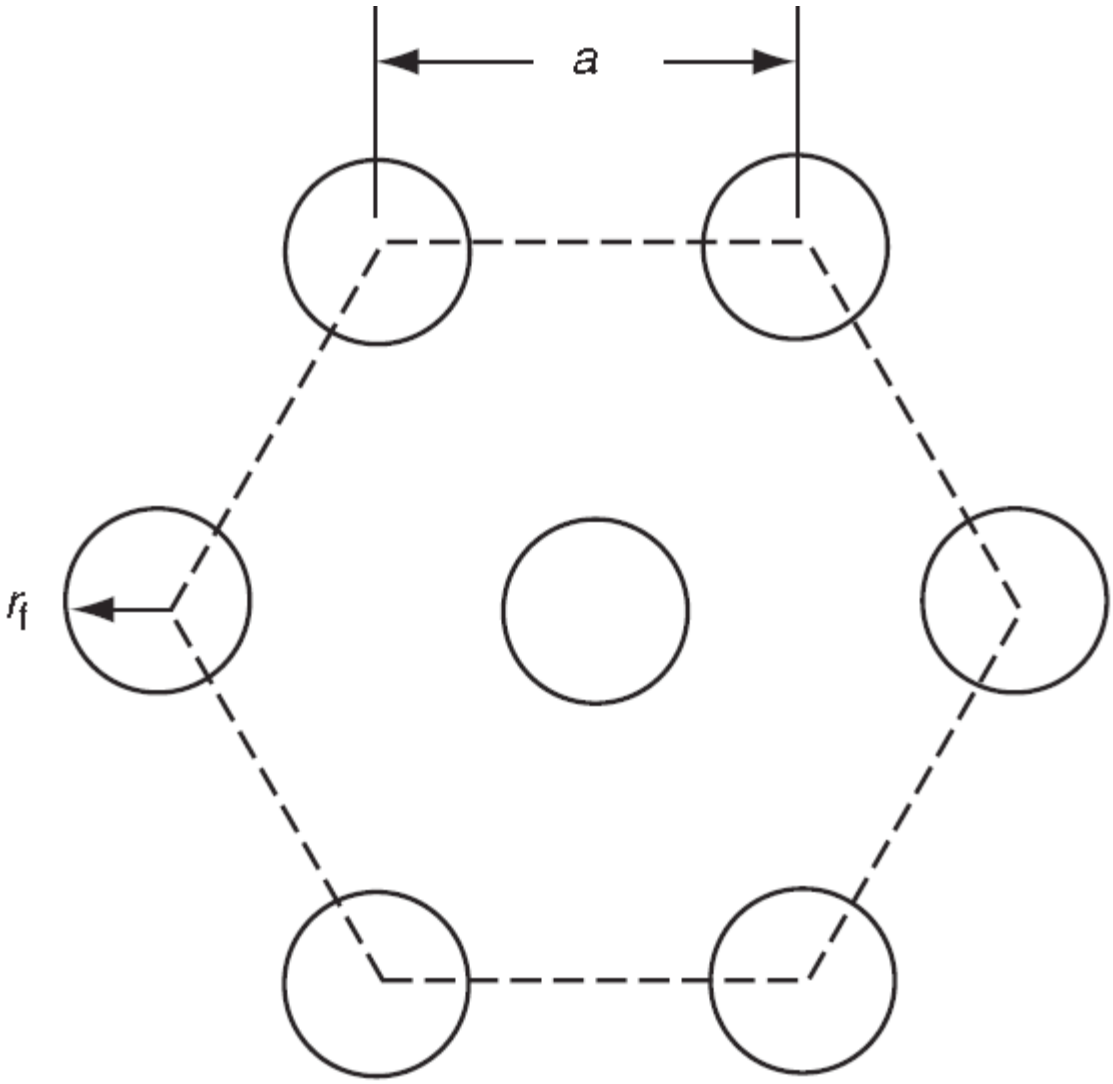


Figure 1.

2A) Describe the specific advantages and limitations of carbon fibre composites when used in the aircraft and automotive industries. (3)

2B) Derive an expression for the composite rigidity modulus of a unidirectional continuous fibre lamina when subjected to pure shear stress. (3)

2C) The material in a composite beam is AS-1 carbon fiber-reinforced epoxy (v_f=0.60). In order to save cost, carbon fibers are being replaced by equal volume percent of E-glass fibers. To compensate for the lower modulus of E-glass fibers, the thickness of the beam is increased threefold. Assuming that the costs of carbon fibers, E-glass fibers, and epoxy are \$40/kg, \$4/kg, and \$6/kg, respectively, determine the percent material cost saved. The density of epoxy is 1.25 g/cm^3. Given, $\rho_{E\text{-}glass}$= 2540 kg/m^3, $\rho_{AS\text{-}1}$= 1800 kg/m^3 and ρ_{epoxy}=1250 kg/m^3 (4)

3A) Illustrate, the step-by-step process of manufacturing of composite materials using (3)

vacuum bagging method.

3B) Derive an expression for the critical fibre volume fraction in a unidirectional continuous fibre composite. (3)

3C) Determine the elements in the stiffness matrix for an angle-ply lamina containing 60 % volume of T-300 carbon fibers in an epoxy matrix. Consider fiber orientation angle 45° for the fiber, $E_f = 220$ GPa and $\nu_f = 0.2$, and for the matrix, $E_m = 3.6$ GPa and $\nu_m = 0.35$. (4)

4A) Describe how a quasi-isotropic layup differs from unidirectional or cross-ply laminates in terms of performance. (3)

4B) A cross ply laminate $[0/90]_s$ made from high strength carbon/epoxy plies and subjected to longitudinal normal force of N_x=100 N/mm. The material properties are E_1=140 GPa, E_2=10GPa, E_6=5GPa, ν_{12}=0.3. Determine the [A] matrix. Assume thickness of each layer as 0.125 mm. (3)

4C) A $[0/90]_s$ symmetric ply laminate is subjected to N_x=100 MPa-mm. Evaluate the resulting stresses in the lamina in the principal material direction. Assume thickness of each layer as 0.125 mm. Material properties of the ply are same as given in Question number 4B. (4)

5A) Describe how cost and processing factors influence the selection of thermoset materials in composite manufacturing. (3)

5B) How do failure theories help analyse and evaluate failure modes in composites, and how does this influence design and performance? (3)

5C) An angle ply lamina made of S-glass/epoxy has the following properties in the principal fibre direction F_{1T} =1280 MPa, F_{1C} = 622 MPa, F_{2T} = 49 MPa, F_{2C} = 245 MPa, F_6 = 69 MPa, E_1 =35 GPa, E_2 = 7 GPa, E_6 =3 GPa, ν_{12} = 0.3. The loading is σ_x = -2 MPa applied at an angle of 60° to the principal fibre direction, σ_y = 4 MPa, and σ_s = -1 MPa. Check the safety of the laminate as per the different failure theories. (4)

-----End-----